

Generalized Probabilistic Topological Field Theories: Notes

Pedro Lauand^{1,*}

¹Perimeter Institute for Theoretical Physics, Waterloo, Ontario, Canada, N2L 2Y5

INTRODUCTION

THE CATEGORY OF GPTS

Generalized probabilistic theories (GPTs) capture the minimal structure a physical theory needs to make probabilistic predictions [1–4]; classical probability and quantum theory are special cases. The theme of this section: that structure is linear algebra plus positivity. A system is a real vector space with a cone and a normalization functional, a transformation is a linear map preserving them, and composites are built on the tensor product. Every definition below is a **Vec** datum plus a positivity condition. We follow Müller [3] and Plávala [4]; for the categorical viewpoint see [5, 8].

Objects: systems

A *state* is an equivalence class of preparation procedures: two preparations are identified when no measurement distinguishes their statistics [3]. Mixing makes the set of states convex, hence embeddable in a real vector space on which outcome probabilities are linear functionals; closure under limits and boundedness of probabilities make it compact; filtering (success probability $\lambda \leq 1$) closes it under nonnegative rescaling. The primitive carrier is thus the cone of *unnormalized* states; normalization survives as a single functional. Throughout, vector spaces are finite-dimensional, real, and spanned by the states they carry.

A *proper cone* $C \subseteq V$ is closed, convex, stable under nonnegative scaling, pointed ($C \cap (-C) = \{0\}$), and generating ($\text{span} C = V$): in order, closure of the state space, mixing and filtering, no vector physical together with its negative, no directions unrelated to physics.

Definition (System). A system is a triple $A = (V_A, V_A^+, u_A)$: a finite-dimensional real vector space V_A , a proper cone $V_A^+ \subseteq V_A$ of unnormalized states, and a unit $u_A \in V_A^*$ with $u_A(v) > 0$ for all $v \in V_A^+ \setminus \{0\}$.

The unit grades the cone by total weight: $u_A(v)$ is the probability that the preparation v succeeds. Normalization is therefore implicit — the *subnormalized* states are $\{v \in V_A^+ : u_A(v) \leq 1\}$, the *normalized* ones the slice

$$\Omega_A := \{\omega \in V_A^+ : u_A(\omega) = 1\},$$

a compact convex base of the cone: every $v \in V_A^+ \setminus \{0\}$ is uniquely $u_A(v)\omega$ with $\omega \in \Omega_A$, strict positivity of u_A being exactly what lets every unnormalized state be normalized. *Pure* states are the extreme points of Ω_A . Alternatively, take (V_A, Ω_A) as primitive and recover cone and unit [3].

Effects. Measurement outcomes need no new data: outcome probabilities are linear in the state, so outcomes are elements of V_A^* and the probability rule is the evaluation pairing of **Vec**. The *effect cone* is the dual cone

$$(V_A^+)^* := \{e \in V_A^* : e(v) \geq 0 \ \forall v \in V_A^+\},$$

the *allowed effects* are $\mathcal{E}_A := [0, u_A]$, so that $e(\omega) \in [0, 1]$ on Ω_A , and a *measurement* is a finite family $(e_i) \subset \mathcal{E}_A$ with $\sum_i e_i = u_A$: outcome probabilities $e_i(\omega)$ are nonnegative and sum to one. By the bipolar theorem $((V_A^+)^*)^* = V_A^+$: states are exactly the vectors nonnegative against all effects. Presentations treating states and effects as co-equal primitives coupled by an abstract pairing reduce to this one once effects are required to separate states and conversely.

No-restriction. Setting $\mathcal{E}_A = [0, u_A]$ declares every admissible effect physical: the *no-restriction hypothesis* [5], our default. Dropping it [6] adds one datum, an effect cone $W_A^+ \subseteq (V_A^+)^*$ spanning V_A^* and containing u_A ; everything below goes through with $(V_A^+)^*$ replaced by W_A^+ .

Examples.

- *Classical:* $V = \mathbb{R}^N$, $V^+ = \mathbb{R}_{\geq 0}^N$, $u(p) = \sum_i p_i$. Ω is the probability simplex, its N vertices the deterministic distributions; effects are the vectors in $[0, 1]^N$. The orthant is self-dual.
- *Quantum:* $V = H_N(\mathbb{C})$, the Hermitian matrices — a real vector space of dimension N^2 — with the PSD cone and $u = \text{tr}$. Ω is the set of density matrices and, via the Hilbert–Schmidt pairing, $\mathcal{E}_A$ the POVM elements $0 \leq E \leq 1$. The PSD cone is self-dual.

- *Boxworld (gbit)*: $V = \mathbb{R}^3$, V^+ the cone over a square [2]. The dual cone is the cone over the *rotated* square, and no linear map aligns the two: not self-dual [3, 10].

Self-duality of the state cone will decide the 1d GPTFT problem.

Morphisms: transformations

Mixing forces transformations to be linear [3]; the rest is preservation of the decorations.

Definition (Transformation). A morphism $f: A \rightarrow B$ is a linear map $f: V_A \rightarrow V_B$ with $f(V_A^+) \subseteq V_B^+$. It is allowed if $u_B \circ f \leq u_A$ on V_A^+ — equivalently, f sends subnormalized states to subnormalized states, the deficit being the failure probability — and a channel if equality holds. Composition and identities are those of $\mathbf{Vec}$.

The effect side takes care of itself: the transpose of a positive map is positive for the dual cones, since $(f^*e)(v) = e(f(v)) \geq 0$. With restricted effects, $f^*(W_B^+) \subseteq W_A^+$ is an independent condition and part of the definition.

$\text{Hom}(A, B)$ is closed under conic combinations but not subtraction: a cone inside the vector space $\text{Hom}_{\mathbf{Vec}}(V_A, V_B)$. GPT thus sits over $\mathbf{Vec}$ as a *wide, non-full* subcategory: all objects, only the positive maps.

The trivial system is $\mathbf{1} = (\mathbb{R}, \mathbb{R}_{\geq 0}, \text{id})$. A morphism $\mathbf{1} \rightarrow A$ is an element of V_A^+ , allowed iff subnormalized; a morphism $A \rightarrow \mathbf{1}$ is an element of $(V_A^+)^*$, allowed iff in $\mathcal{E}_A$. States and effects are ordinary arrows into and out of the unit. In particular u_A is itself a morphism $A \rightarrow \mathbf{1}$, and channels are exactly the morphisms over $\mathbf{1}$: $u_B \circ f = u_A$.

Composites: the monoidal structure

Independent local preparations and measurements must combine into joint ones with factorizing statistics,

$$(e_A \otimes e_B)(\omega_A \otimes \omega_B) = e_A(\omega_A)e_B(\omega_B). \quad (1)$$

Mixing makes both products bilinear, so the joint space contains $V_A \otimes V_B$, with product states and effects the ordinary tensors [3]. Whether it is exactly $V_A \otimes V_B$ is a postulate:

Definition (Tomographic locality). Joint states are determined by the joint statistics of local measurements [1, 2]; equivalently, $V_{AB} = V_A \otimes V_B$.

We assume it by default. The local data bound the composite cone without fixing it: the *minimal tensor cone*

$$V_A^+ \otimes_{\min} V_B^+ := \text{cone}\{v \otimes v' : v \in V_A^+, v' \in V_B^+\}$$

is generated by the product states, while the *maximal tensor cone*

$$V_A^+ \otimes_{\max} V_B^+ := \{g : (e \otimes e')(g) \geq 0 \ \forall e \in (V_A^+)^*, e' \in (V_B^+)^*\}$$

contains everything consistent with product measurements. Both are proper, and duality exchanges them:

Lemma 1 (Min–max duality [4, 7]). Under $(V_A \otimes V_B)^* \cong V_A^* \otimes V_B^*$, $(C_A \otimes_{\min} C_B)^* = C_A^* \otimes_{\max} C_B^*$ and $(C_A \otimes_{\max} C_B)^* = C_A^* \otimes_{\min} C_B^*$.

Definition (Composite). A composite of A and B is a system $AB = (V_A \otimes V_B, V_{AB}^+, u_A \otimes u_B)$ with

$$V_A^+ \otimes_{\min} V_B^+ \subseteq V_{AB}^+ \subseteq V_A^+ \otimes_{\max} V_B^+. \quad (2)$$

Any intermediate proper cone is admissible: a GPT includes a coherent (associative, symmetric) choice of composites, and any such choice makes $(\text{GPT}, \otimes, \mathbf{1})$ symmetric monoidal. The freedom in (2) is generic:

Theorem 1 (Entangleability of cones [7]). $C_A \otimes_{\min} C_B = C_A \otimes_{\max} C_B$ iff C_A or C_B is simplicial (classical).

Any two nonclassical systems therefore admit entangled states or effects. For quantum systems the three cones in (2) all differ: separable $\subsetneq$ PSD $\subsetneq$ entanglement witnesses. The quantum tensor product is an additional input — a choice strictly between the bounds [3].

No-signalling holds automatically: local effects sum to the local unit, so marginals on A do not depend on the measurement chosen on B ; the *reduced state* is $\omega_A := (\text{id}_A \otimes u_B)(\omega_{AB})$ [3].

Beyond tomographic locality. Without the postulate, the tensor product still embeds: $V_{AB} \cong (V_A \otimes V_B) \oplus C$, with a *global sector* C invisible to local tomography; $C=0$ iff the composite is tomographically local. Real-vector-space quantum theory [9] and fermionic systems (parity superselection) have $C \neq 0$. The GPTFT constructions below do not require $C=0$.

GPT versus Vec

Vec. An object is a finite-dimensional real vector space V ; a morphism is any linear map, so $\text{Hom}(V_A, V_B)$ is itself a vector space; the monoidal product is $\otimes$, with unit $\mathbb{R}$. The category is compact closed: every object has a dual V^* , the evaluation $\text{ev}(\varphi \otimes v) = \varphi(v)$ and coevaluation $\text{coev}(1) = \sum_a f_a \otimes f^a$ (dual bases) are canonical, and the snake identities hold for every V — there is never anything to check.

GPT. The same three layers, each carrying one positivity constraint (forgetting them is a functor back to **Vec**):

$$\begin{aligned} V &\rightsquigarrow (V, V^+, u), \\ f &\rightsquigarrow f \text{ with } f(V_A^+) \subseteq V_B^+, \\ V_A \otimes V_B &\rightsquigarrow V_{AB}^+ \in [\otimes_{\min}, \otimes_{\max}], \quad u_A \otimes u_B. \end{aligned}$$

Each constraint changes the linear-algebraic character of its layer. The hom-set

$$\text{Hom}_{\text{GPT}}(A, B) = \{f \in \text{Hom}_{\text{Vec}}(V_A, V_B) : f(V_A^+) \subseteq V_B^+\}$$

is a cone, not a subspace: positivity of f does not give positivity of $-f$. The monoidal product is no longer a single functor but a family, the underlying space and unit fixed, the composite cone a choice in (2). Duals survive at the level of objects — swapping the state cone with the effect cone gives a dual system on V^* — but ev and coev are morphisms only if *positive*: the element $\sum_a f_a \otimes f^a$ must lie in the chosen composite cone, and in the dual of the cone chosen for the swapped pair. In **Vec** dualizability is automatic; in GPT it is a contingent property of the cones. Which GPTs pass is the subject of the next section.

1D TFTS

1D TQFTs

1D GPTFTs

* plauand@perimeterinstitute.ca

- [1] L. Hardy, *Quantum theory from five reasonable axioms*, arXiv:quant-ph/0101012.
- [2] J. Barrett, *Information processing in generalized probabilistic theories*, Phys. Rev. A **75**, 032304 (2007), arXiv:quant-ph/0508211.
- [3] M. P. Müller, *Probabilistic theories and reconstructions of quantum theory*, SciPost Phys. Lect. Notes **28** (2021), arXiv:2011.01286.
- [4] M. Plávala, *General probabilistic theories: An introduction*, Phys. Rep. **1033**, 1 (2023), arXiv:2103.07469.
- [5] G. Chiribella, G. M. D'Ariano, and P. Perinotti, *Probabilistic theories with purification*, Phys. Rev. A **81**, 062348 (2010), arXiv:0908.1583.
- [6] P. Janotta and R. Lal, *Generalized probabilistic theories without the no-restriction hypothesis*, Phys. Rev. A **87**, 052131 (2013), arXiv:1302.2632.
- [7] G. Aubrun, L. Lami, C. Palazuelos, and M. Plávala, *Entangleability of cones*, Geom. Funct. Anal. **31**, 181 (2021), arXiv:1911.09663.
- [8] S. Gogioso and C. M. Scandolo, *Categorical probabilistic theories*, EPTCS **266**, 367 (2018), arXiv:1701.08075.
- [9] L. Hardy and W. K. Wootters, *Limited holism and real-vector-space quantum theory*, Found. Phys. **42**, 454 (2012), arXiv:1005.4870.
- [10] H. Barnum and A. Wilce, *Post-classical probability theory*, in *Quantum Theory: Informational Foundations and Foils*, edited by G. Chiribella and R. W. Spekkens (Springer, Dordrecht, 2016), arXiv:1205.3833.